IAT New Test 15

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

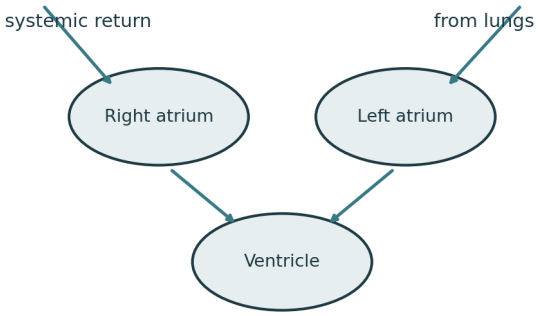
Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 A fern produces two visibly different spore sizes, and the large spore develops into a female gametophyte. Which inference is warranted?
A) All fern spores are diploid
B) This fern is heterosporous, with megaspores giving female gametophytes
C) The small spores become sporophytes directly
D) The large spores are seeds enclosed by fruit

Q2 The labelled frog heart has two atria draining into one ventricle. Which chamber directly receives oxygenated blood returning from the lungs?

Frog heart: return pathways

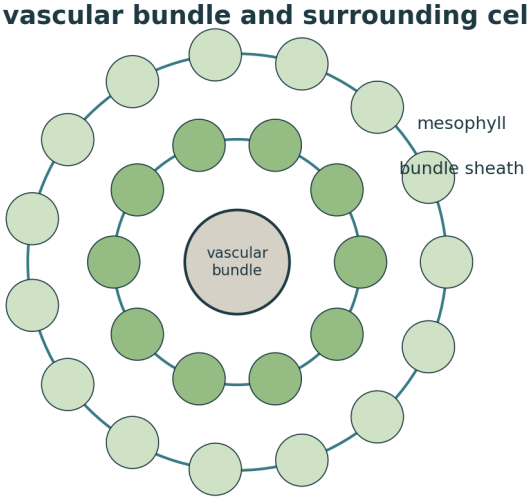


A) Right atrium
B) Ventricle
C) Left atrium
D) Sinus venosus

Q3 Two equal cells are placed in the same hypertonic solution. Cell X loses water and shrinks; cell Y initially shrinks but then returns toward its original volume when a specific solute carrier is activated. Which control most directly tests whether carrier-dependent solute uptake drives Y's recovery?
A) Block the carrier in Y while holding external osmolarity fixed
B) Remove X from the experiment
C) Change both solution osmolarities unpredictably
D) Increase light intensity without monitoring solute

Q4 The leaf section shows a ring of chloroplast-rich bundle-sheath cells around each vascular bundle and an outer mesophyll layer. Which interpretation fits this anatomy and its carbon-fixation role?

Leaf: vascular bundle and surrounding cells



A) Kranz anatomy supports C4 transfer of four-carbon acids to bundle-sheath Calvin-cycle cells
B) It proves photorespiration is impossible at all temperatures
C) It is a nonphotosynthetic stem cross-section
D) It is ordinary C3 anatomy with no bundle-sheath function

Q5 A kidney filters 120 mL plasma per minute, and a freely filtered substance has plasma concentration 2 mg mL^{-1}. Its urine excretion is 60 mg min^{-1} with no secretion. What fraction of filtered load was reabsorbed?
A) 1/4
B) 1/2
C) 3/4
D) 7/8

Q6 A human primary spermatocyte completes meiosis normally. How many functional haploid sperm can ultimately arise from it, assuming all products survive maturation?
A) One
B) Two
C) Three
D) Four

Q7 Two unlinked loci segregate independently in an AaBb × AaBb cross, with complete dominance at each locus. Given that an offspring shows a dominant phenotype at at least one locus, what is the probability it is homozygous recessive at exactly one locus?
A) 2/5
B) 3/8
C) 1/4
D) 1/2

Q8 A tumour sample has a mutation disabling a cell-cycle checkpoint that normally responds to DNA damage. Which pattern is most consistent with loss of that checkpoint?
A) More time for DNA repair before mitosis
B) Cells enter division despite unrepaired lesions
C) No further DNA replication in any cells
D) All cells immediately undergo normal differentiation

Q9 A proposed gene-edited crop resists a fungal infection in greenhouse tests. Which observation most directly tests whether the edit, rather than background variation, caused resistance?
A) Only compare edited plants after spraying different doses
B) Measure leaf colour in one edited plant
C) Compare independently edited lines with isogenic unedited controls under the same inoculum
D) Assume a marker band proves field resistance

Q10 A protected forest and a similarly sized unprotected forest are surveyed for species with equal effort. The protected site has more species now, but it already had more before protection. Which added comparison best tests conservation effect?
A) Current totals only
B) Species names in a single plot
C) Current area ratio only
D) Change in each site's species count from before to after protection

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Q11 An X-linked recessive disorder affects a boy, while his father is unaffected. His mother is unaffected and has one affected brother. Assuming no new mutations, what is the probability that the mother's next son is affected, given she is confirmed a carrier by the affected boy?
A) 1/4
B) 1/2
C) 1/3
D) 1

Q12 At a synapse, excitatory transmitter release and receptor binding are normal, but a second inhibitory input arrives simultaneously. The recorded membrane potential rises by 4 mV with excitation alone and by only 1 mV when both inputs fire. What inference is strongest?
A) Excitatory vesicle fusion was abolished
B) The inhibitory input must be located in the presynaptic cell
C) Postsynaptic summation reduced net depolarisation
D) All receptors became permanently inactive

Q13 After a carbohydrate-rich meal, blood glucose rises in a healthy person. Which paired endocrine response is expected to move glucose toward baseline?
A) Insulin decreases, glucagon rises
B) Insulin rises, glucagon also rises to the same effect
C) Both hormones vanish
D) Insulin rises and glucagon falls

Q14 A plant-cell homogenate yields a low-speed nuclear pellet, then a fraction rich in chlorophyll and thylakoid membranes. Which process is best enriched in the second fraction under illumination?
A) Light-driven electron transport and proton-gradient formation
B) Nuclear DNA replication
C) Glycolysis in cytosol
D) Protein translation solely on free cytosolic ribosomes

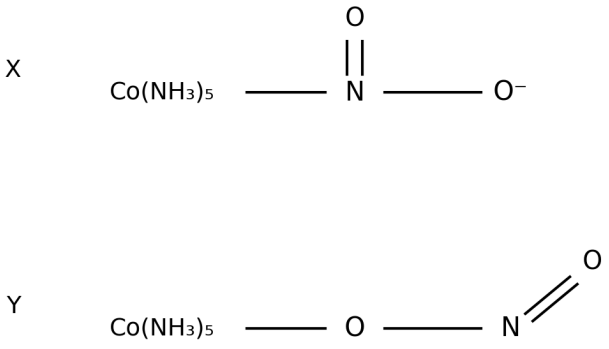
Q15 A normal cycle shows an LH surge followed by ovulation. If no pregnancy occurs, which change helps initiate menstruation later?
A) Sustained high progesterone from placenta
B) Corpus luteum regression lowers progesterone and estrogen
C) Continued LH surge maintains corpus luteum indefinitely
D) Endometrium sheds before ovulation

Chemistry

Q16 Which Lewis contributor for carbonate CO_3^{2-} respects the octet on second-row carbon?
A) Three C=O double bonds
B) One C=O and two C–O⁻ bonds
C) One C≡O and two C–O bonds
D) Carbon bonded to only one oxygen

Q17 The two structures have identical composition $[Co(NH_3)_5(NO_2)]Cl_2$. In X the metal is bonded to nitrogen of the NO_2 ligand; in Y it is bonded to oxygen of that ligand. What kind of isomerism distinguishes them, and what is the coordination number in each?

Nitrite donor-atom connectivity



Both coordination cations have charge +2; each has two external Cl^-

A) Geometrical; five in each
B) Ionisation; six in X and five in Y
C) Linkage; six in each
D) Optical; seven in each

Q18 Hydrogen emits a photon in the transition n=3→n=2. In terms of Rydberg constant R, what is its inverse wavelength 1/λ?
A) R/36
B) R/9
C) R/4
D) 5R/36

Q19 A molecule contains both a nonconjugated C=C double bond and a ketone C=O group. One equivalent of Br_2 is added in dry CCl_4 in the dark. Which electron-pair movement starts the characteristic bromine addition, and what group remains unchanged at this stage?
A) The alkene π pair attacks electrophilic Br; the ketone C=O remains
B) Carbonyl oxygen attacks Br^-; the alkene remains
C) The ketone π pair attacks Br^-; the alkene remains
D) The alkene π pair attacks Br^-; the carbonyl oxygen leaves

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Q20 In peroxide-free HBr addition to propene, which major constitutional product and reason are expected by the usual ionic route?
A) 2-bromopropane via more stable secondary carbocation
B) 1-bromopropane via primary carbocation
C) 1,2-dibromopropane via two bromide ions
D) Propane via reduction

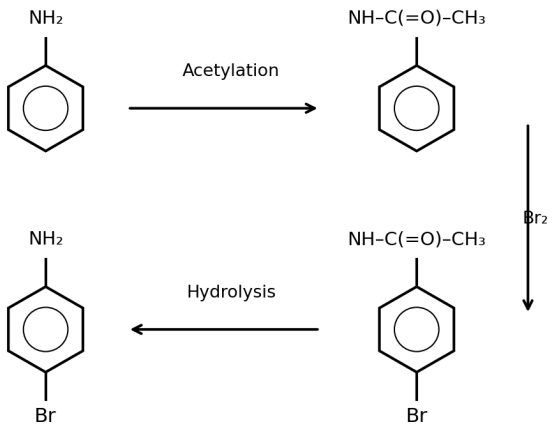
Q21 A tertiary alkyl halide reacts in aqueous ethanol and gives both substitution and elimination products. Which observation most supports a carbocation-mediated SN1 contribution?
A) Complete inversion at one stereocentre only
B) Rate doubles when nucleophile concentration doubles at fixed substrate
C) Rate depends chiefly on substrate concentration while nucleophile concentration changes little
D) Reaction requires a primary carbon only

Q22 An unknown alcohol is oxidised gently: the product gives Tollens positive and can be oxidised further to a carboxylic acid. Which alcohol class best fits?
A) Tertiary
B) Phenol
C) Secondary
D) Primary

Q23 In Fischer esterification of an acid and alcohol, removing water as it forms increases ester yield at unchanged temperature. Why?
A) Removal shifts reversible equilibrium toward ester and water products
B) Water acts as irreversible catalyst only
C) Removal changes the acid into an alkane
D) Alcohol becomes a stronger oxidant

Q24 The structural sequence shows aniline converted to acetanilide, then predominantly para-bromoacetanilide under controlled bromination, then para-bromoaniline by hydrolysis. Why is acetylation used before bromination?

Aniline protection, substitution and deprotection

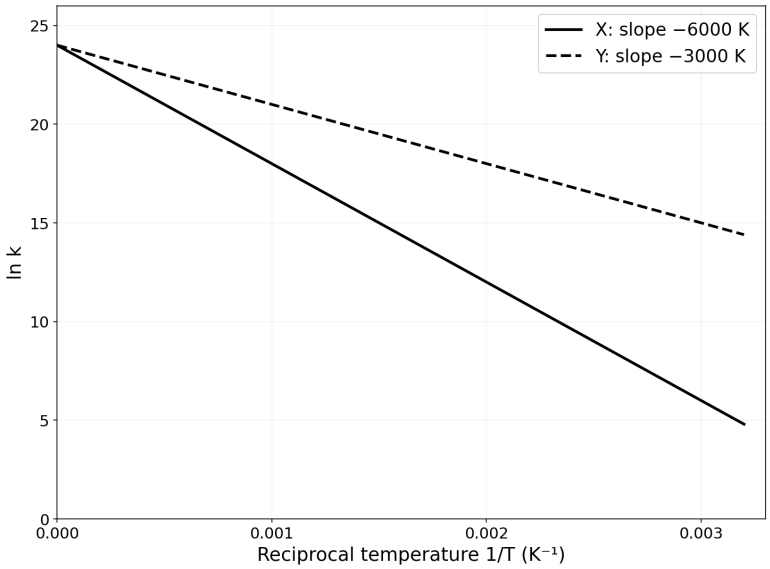


A) To change the amino substituent into a meta director permanently
B) To reduce excessive ring activation while retaining ortho/para direction, then restore NH_2
C) To increase nitrogen donation enough to make tribromination selective for para
D) To replace the amino nitrogen by oxygen before aromatic substitution

Q25 A 0.10 m electrolyte solution has freezing depression 0.28 K in a solvent with Kf=1.40 K kg mol^{-1}. What is the observed van't Hoff factor?
A) 1
B) 1.5
C) 2
D) 3

Q26 A concentration cell uses identical metal electrodes with M^{2+} activities 0.01 and 1.00 at the same temperature. Which side acts as anode spontaneously?
A) Concentrated side, to consume ions
B) Neither, since standard potentials cancel
C) Both simultaneously
D) Dilute side, generating ions to reduce the concentration difference

Q27 Two Arrhenius plots ln k versus 1/T for reactions X and Y are straight; X has slope −6000 K and Y slope −3000 K. If both have the same pre-exponential factor, which has larger activation energy?



A) X, twice Y
B) Y, twice X
C) Equal energies
D) Cannot infer from slopes

Q28 A rigid bomb calorimeter is calibrated by supplying 1000 J electrically; its temperature rises by 2.00 K. Combustion of 0.0100 mol of a fuel then raises its temperature by 3.00 K. For the stated combustion reaction, gas mole change is $\Delta n_g = -1$ per mole fuel. At 300 K, with $R = 8.30\,\mathrm{J\,mol^{-1}K^{-1}}$, estimate the reaction's molar enthalpy change.
A) −150.0 kJ mol^{-1}
B) −152.5 kJ mol^{-1}
C) +150.0 kJ mol^{-1}
D) −147.5 kJ mol^{-1}

Q29 A weak monoprotic acid HA has initial concentration 0.10 M and Ka=10^{-5}. Assuming small dissociation, estimate its degree of ionisation α.
A) 10^{-3}
B) 10^{-1}
C) 10^{-2}
D) 10^{-4}

Q30 Under acidic conditions, MnO_4^- is a stronger oxidant than under strongly basic conditions. Which change in reduction product is consistent with the usual qualitative trend?
A) Mn(VII) never changes
B) Only oxygen is reduced
C) Mn becomes +8 in acid
D) Acidic reduction can reach Mn^{2+} whereas basic reduction often stops at MnO_2 or MnO_4^{2-}

Mathematics

Q31 A circle centred on the x-axis passes through (0,0) and (6,0). The line x=1 cuts it at P and Q. Tangents at P and Q meet at T. What is the distance OT from the origin?
A) 9/2
B) 3
C) 6
D) 3/2

Q32 An ellipse centred at origin has foci (±3,0) and directrices x=±25/3. What is its equation?
A) $x^2/25+y^2/16=1$
B) $x^2/16+y^2/25=1$
C) $x^2/9+y^2/16=1$
D) $x^2/25+y^2/9=1$

Q33 Find the area enclosed by $y=4-x^2$ and y=0 between their two x-intercepts.
A) 16/3
B) 32/3
C) 8
D) 4

Q34 Evaluate $\int_1^e (\ln x)/x\,dx$.
A) 1
B) 1/4
C) 1/2
D) 2

Q35 For f(x)=x+9/x on x>0, which interval is f strictly decreasing?
A) (3,∞)
B) (0,∞)
C) (0,9)
D) (0,3)

Q36 Let $f:(0,\infty)\to[2,\infty)$ be f(x)=x+1/x. Which classification is correct?
A) Surjective but not injective
B) Injective but not surjective
C) Both injective and surjective
D) Neither

Q37 For an invertible 3×3 matrix A with determinant 2, what is determinant of adj(A)?
A) 2
B) 4
C) 8
D) 1/2

Q38 Events A and B have P(A)=1/2, P(B)=1/2 and are mutually exclusive. Can they also be independent?
A) Yes, always
B) Only if both are impossible
C) No, since $P(A\cap B)=0$ differs from $P(A)P(B)=1/4$
D) Only if sample space has two outcomes

Q39 How many distinct x in [0,2π) solve cos(3x)=1?
A) 2
B) 4
C) 6
D) 3

Q40 A line through (1,0,0) has direction vector (1,k,1) and is parallel to the plane $2x - y + z = 5$. What are k and the perpendicular distance between the line and the plane?
A) $3;\ \sqrt{6}/2$
B) $3;\ 3$
C) $-3;\ \sqrt{6}/2$
D) $1;\ \sqrt{6}$

Q41 Points A=(0,1,2), B=(2,3,4), C=(5,6,7) are given. Are they collinear, and what is ratio AB:BC of their segment lengths?
A) No; 2:3
B) Yes; 2:3
C) Yes; 3:2
D) No; 1:1

Q42 For $z = x + iy$, the locus $|z - 1| = 2|z + 1|$ is a circle. What are its centre and radius?
A) $(5/3, 0);\ 4/3$
B) $(-5/3, 0);\ 2/3$
C) $(-5/3, 0);\ 4/3$
D) $(0, 0);\ 1$

Q43 For which real k does $f(x) = x^2 - 2kx + 2$ have one root strictly in (0,1) and the other strictly in (2,3)?
A) $[3/2, 11/6]$
B) $(3/2, \infty)$
C) $(-\infty, 11/6)$
D) $(3/2, 11/6)$

Q44 In expansion $(1+x)^8$, what is the sum of coefficients of x^3 and x^5?
A) 112
B) 56
C) 70
D) 28

Q45 Three arithmetic means are inserted between 4 and 16 to form a five-term arithmetic progression. What is the sum of the three inserted means?
A) 20
B) 30
C) 40
D) 60

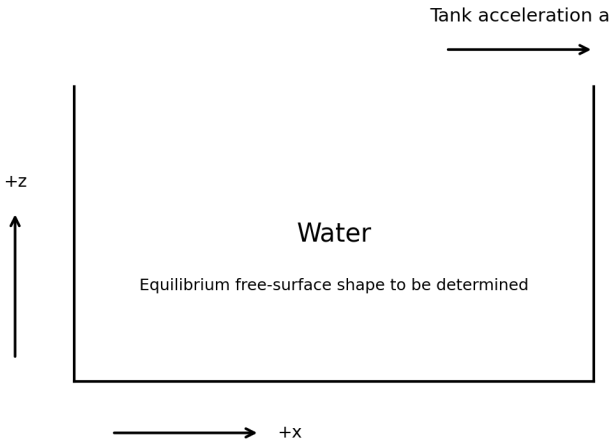
Physics

Q46 Two cars start together at x=0. A travels at constant +12 m s^{-1} and B starts from rest with acceleration +3 m s^{-2}. When do they next have the same position?
A) 4 s
B) 6 s
C) 8 s
D) 12 s

Q47 A rectangular sheet has measured sides (10.0±0.1) cm and (20.0±0.2) cm. Using maximum fractional-error propagation, what is approximate maximum percentage error in area?
A) 1%
B) 1.5%
C) 4%
D) 2%

Q48 An open tank of water accelerates horizontally to the right with acceleration a. The diagram specifies the tank axes and acceleration but not the final water surface. In the tank frame, which side is higher at equilibrium, and what is the tangent of the surface's angle to the horizontal?

Tank frame: final surface is not supplied



A) Left; a/g
B) Right; a/g
C) Left; g/a
D) Right; g/a

Q49 A uniform disk of mass M, radius R has a small concentric circular hole of radius R/2 removed. What is the moment of inertia of the remaining annulus about its central perpendicular axis in terms of its remaining mass m?
A) $mR^2/2$
B) $5mR^2/8$
C) $3mR^2/4$
D) mR^2

Q50 A cyclic heat engine absorbs 800 J per cycle and converts 40% of it to shaft work. All its rejected heat feeds a second cyclic engine of efficiency 1/3. A parasitic external load consumes 40 J from the combined shaft output per cycle, without returning heat to either engine. What are the useful efficiency relative to the original 800 J input and the heat rejected by the second engine?
A) 60%; 320 J
B) 55%; 480 J
C) 55%; 320 J
D) 40%; 480 J

Q51 Two springs of constants k and 3k support a mass m in parallel. What is oscillation period divided by period with spring k alone?
A) 2
B) 1/2
C) 1/4
D) $\sqrt{3}/2$

Q52 Two coherent sources have equal amplitudes and a fixed initial phase difference π at a screen point where their path lengths are equal. What intensity relative to the one-source intensity I_0 results there?
A) $4I_0$
B) $2I_0$
C) 0
D) I_0

Q53 A dipole of moment p in uniform electric field E is rotated from parallel to perpendicular without changing p or E. What is increase in potential energy?
A) $-pE$
B) 0
C) 2pE
D) $+pE$

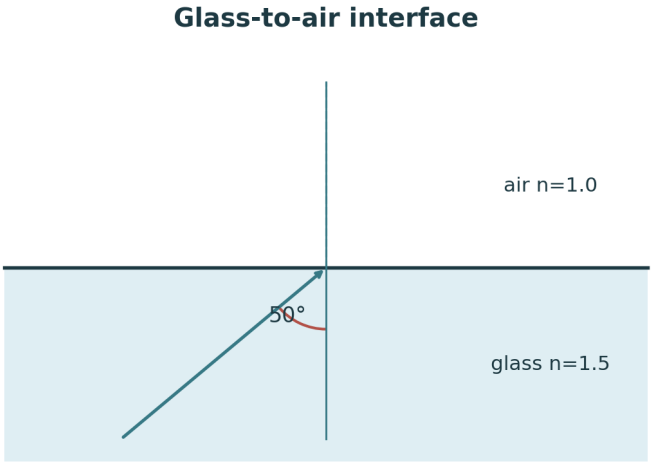
Q54 A wire carries 4 A through cross-sectional area 2 mm^2. What is current density magnitude?
A) 2×10^{6} A m^{-2}
B) 2 A m^{-2}
C) 8×10^{-6} A m^{-2}
D) 8×10^{6} A m^{-2}

Q55 A closed sample initially contains 900 radioactive parent atoms of types P and Q in total, with equal initial activities. Their half-lives are 2 h and 4 h. Each parent produces one stable daughter, initially absent. What is the expected total number of daughter atoms after 4 h?
A) 375
B) 450
C) 600
D) 525

Q56 Two coaxial loops have defined positive normal toward the observer. Current in primary rises and its field through secondary points toward observer. What induced field direction does secondary current create?
A) Toward observer, aiding change
B) Away from observer, opposing increasing flux
C) Zero because loops do not touch
D) Alternates instantaneously with no change in primary direction

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Q57 A ray goes from glass n=1.5 into air n=1.0 at incidence angle 50° measured to normal. What happens at the interface?



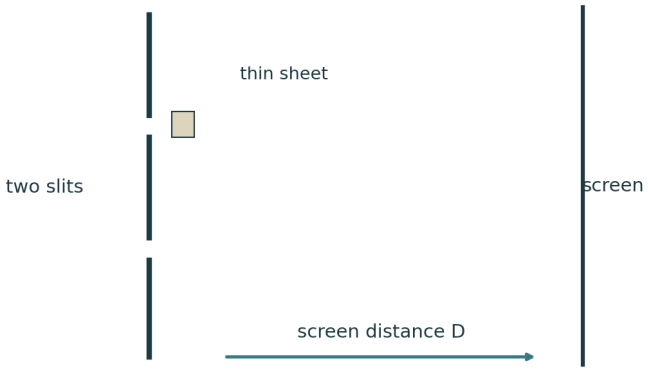
A) Total internal reflection
B) Refraction into air at 50°
C) Refraction into air at 30°
D) Absorption of all light

Q58 A series RLC circuit has R=3 Ω and net reactance XL−XC=4 Ω. What is phase angle φ between source voltage and current, expressed as tanφ?
A) 3/4
B) 4/3
C) 7/3
D) 1/7

Q59 A half-wave rectifier fed by $v(t)=V_0 \sin \omega t$ drives resistance R with ideal diode. What is average output power over a full cycle?
A) V_0^2/R
B) $V_0^2/(2R)$
C) $V_0^2/(4R)$
D) $V_0^2/(8R)$

Q60 In a Young setup the original fringe width is $\beta = \lambda D/d$. The slit separation is changed to 2d, and a thin sheet of refractive index 1.50 is placed at one slit. The tracked bright fringe with zero total optical path difference shifts by $3\beta/4$ from the geometric axis. In the paraxial approximation, what are the new fringe width and sheet thickness?

Two-slit geometry with a thin sheet



A) $2\beta;\ 3\lambda$
B) $\beta/2;\ 3\lambda/2$
C) $\beta/2;\ 6\lambda$
D) $\beta/2;\ 3\lambda$

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / B pteridophyte spore type

Answer: (B) Heterospory.

Why: Distinct microspores and megaspores indicate heterospory, and the larger megaspore develops into the female gametophyte. A spore remains a haploid meiotic product; this observation does not mean a seed or fruit has formed.

Why not the others: (A) ignores meiosis; (C) skips the gametophyte generation; (D) assigns angiosperm fruit to a fern.

Key point: Separate heterospory from seed production.

Q2 / C frog circulatory structure

Answer: (C) Left atrium.

Why: Pulmonary venous return enters the frog's left atrium, whereas the right side receives systemic venous return through the sinus venosus. The single ventricle receives blood from both atria after their inflow; it is not the first chamber for pulmonary return.

Why not the others: (A) receives systemic blood; (B) is downstream; (D) channels systemic return.

Key point: Follow the labelled inflow before ventricular mixing.

Q3 / A membrane transport control

Answer: (A) Block carrier under matched osmotic conditions.

Why: If Y's recovery depends on importing solute, blocking that route should prevent restoration of intracellular osmolarity and consequent water entry. Holding external osmolarity fixed isolates transport from a changed water gradient.

Why not the others: (B) removes comparison; (C) confounds osmotic drive; (D) changes an irrelevant variable.

Key point: Alter one causal transport step at a time.

Q4 / A C3 versus C4 anatomy

Answer: (A) Kranz anatomy with spatially separated fixation.

Why: The concentric, chloroplast-bearing bundle sheath is characteristic of C4 Kranz anatomy and spatial separation of initial mesophyll fixation from the Calvin cycle. This raises local CO_2 around RuBisCO but does not logically make photorespiration impossible in every condition.

Why not the others: (B) overstates suppression; (C) ignores leaf features; (D) ignores the distinctive bundle sheath.

Key point: Link anatomy to spatial separation of fixation.

Q5 / C nephron clearance and reabsorption

Answer: (C) 3/4.

Given: Filtered load=120×2=240 mg/min.

Formula/Principle: With no secretion, reabsorbed=filtered−excreted=180 mg/min.

Step 1: 180/240=3/4.

Why: Clearance and filtration cannot be equated when tubular reabsorption occurs; urine contains only the unreabsorbed part under the stated no-secretion condition. The ratio uses mass rates in matching units.

Why not the others: (A) is excreted fraction; (B) halves arbitrarily; (D) undercounts excretion.

Key point: Compare filtered and excreted mass per time.

Q6 / D human gametogenesis

Answer: (D) Four.

Why: Meiosis I creates two secondary spermatocytes, and meiosis II creates four spermatids; spermiogenesis can mature each into a sperm without additional chromosome divisions. This differs from oogenesis, where cytokinesis yields one large ovum and polar bodies.

Why not the others: (A) applies oogenesis output; (B) stops after meiosis I; (C) loses one daughter without premise.

Key point: Count meiotic products before maturation.

Q7 / A Conditioned two-locus phenotype classes

Answer: (A) 2/5.

Given: Each locus gives dominant phenotype with probability $3/4$ and recessive phenotype with probability $1/4$.

Formula/Principle: Count the two mutually exclusive exactly-one-recessive classes, then condition on not being recessive at both loci.

Step 1:

$$P(aaB_) = P(A_bb) = \frac{1}{4}\frac{3}{4} = \frac{3}{16}.$$

Step 2:

$$P(\text{exactly one recessive locus}) = \frac{6}{16}.$$

Step 3:

$$P(\text{at least one dominant locus}) = 1 - P(aabb) = 1 - \frac{1}{16} = \frac{15}{16}.$$

Step 4:

$$P = \frac{6/16}{15/16} = \frac{2}{5}.$$

Why: Independent segregation applies before conditioning; the observed phenotype restricts the joint sample space. Both one-recessive classes satisfy that condition, whereas aabb does not.

Why not the others: (B) is the unconditioned $6/16$; (C) uses a single-locus recessive probability; (D) treats the permitted joint phenotypes as equally likely.

Key point: Distinguish exactly one, at least one and both when combining loci.

Q8 / B cancer and cell-cycle link

Answer: (B) Premature progression.

Why: A defective damage checkpoint permits progression while lesions persist, allowing mutations to accumulate through later divisions. It does not itself guarantee tumour formation, but it removes one restraint on proliferation.

Why not the others: (A) is intact arrest; (C) is unrelated complete block; (D) is not caused by checkpoint loss.

Key point: Identify the safeguard removed by mutation.

Q9 / C biotechnology application evidence

Answer: (C) Isogenic controlled comparison.

Why: Matching genetic background and exposure isolates the introduced edit, while independent lines reduce the chance that an unrelated insertion or culture effect explains resistance. A marker confirms molecular presence, not the causal field phenotype.

Why not the others: (A) confounds exposure; (B) misses infection; (D) substitutes marker for phenotypic test.

Key point: Use matched controls and independent replicates.

Q10 / D biodiversity conservation evidence

Answer: (D) Compare changes over time.

Why: A difference in current totals cannot identify an effect when the baseline sites differed. Comparing within-site changes and then between-site changes controls the initial gap more directly, though other contemporaneous factors still require review.

Why not the others: (A) repeats confounded comparison; (B) lacks baseline; (C) ignores species trend.

Key point: Establish baseline before attributing conservation effects.

Q11 / B conditional pedigree risk

Answer: (B) 1/2.

Given: The affected boy received mutant X from mother, establishing her heterozygosity under assumptions.

Formula/Principle: Each son gets maternal X independently and paternal Y.

Step 1: Carrier mother transmits mutant X with probability 1/2 to next son.

Why: The affected brother in the maternal family supports carrier possibility, but the existing affected son is the decisive information under the no-new-mutation premise. A new pregnancy has independent segregation; prior sons do not deplete the allele.

Why not the others: (A) combines sex probability though next child is given son; (C) uses conditional carrier uncertainty already resolved; (D) treats transmission as certain.

Key point: Condition on known carrier status and son’s sex.

Q12 / C neural synapse and receptor

Answer: (C) Summation reduces net depolarisation.

Why: The 4 mV response verifies the excitatory pathway, and the smaller combined response supports an opposing inhibitory postsynaptic contribution. It does not reveal an exact ion-channel identity, but it distinguishes integration of two inputs from failure of transmitter release.

Why not the others: (A) contradicts excitation-alone record; (B) places an unobserved location; (D) contradicts retained response.

Key point: Compare isolated and combined synaptic inputs.

Q13 / D endocrine feedback

Answer: (D) Insulin up, glucagon down.

Why: Higher glucose promotes pancreatic beta-cell insulin secretion and suppresses alpha-cell glucagon drive under the ordinary fed response. Insulin favours uptake and storage, while lower glucagon reduces hepatic glucose release; the two changes jointly oppose hyperglycaemia.

Why not the others: (A) favours glucose release; (B) pairs opposing actions; (C) is not physiological feedback.

Key point: Trace hormone effects on circulating glucose.

Q14 / A organelle fractionation

Answer: (A) Thylakoid light reactions.

Why: Chlorophyll and thylakoid membranes identify chloroplast material whose photosystems perform light-driven electron transport. Enrichment is relative, not absolute; separating nuclei first does not place nuclear or soluble cytosolic functions in the chloroplast fraction.

Why not the others: (B) belongs nuclear pellet; (C) is cytosolic; (D) uses free ribosomes outside the fraction.

Key point: Infer activity from specific fraction markers.

Q15 / B menstrual hormone sequence

Answer: (B) Luteal hormones fall.

Why: Without embryonic hCG support, the corpus luteum regresses and its progesterone/estrogen output falls. Withdrawal of hormonal support causes the endometrium to break down after the luteal phase, not at the pre-ovulatory surge.

Why not the others: (A) needs pregnancy; (C) ignores transient surge; (D) reverses cycle order.

Key point: Link luteal regression to hormone withdrawal.

Chemistry

Q16 / B resonance contributor validity

Answer: (B) One double, two single.

Why: Central carbon has four shared electron pairs and eight surrounding electrons in the valid contributor; two oxygens carry -1 formal charge. Resonance distributes the double-bond position among three equivalent structures without giving carbon an expanded octet.

Why not the others: (A) gives carbon 12 electrons; (C) gives five bonding pairs; (D) omits two bonded oxygens.

Key point: Apply the octet before judging resonance.

Q17 / C Ambidentate donor connectivity and linkage isomerism

Answer: (C) Linkage isomerism; coordination number six in each.

Why: Nitrite is an ambidentate ligand: it can donate through N or through O while retaining the same chemical composition. Switching the donor atom changes linkage, not the number of occupied sites; five ammonia donors and one nitrite donor still give six. Both chloride counterions remain outside the same coordination sphere.

Step 1: Read the metal–N versus metal–O connectivity in the diagram.

Step 2: Count $5 + 1 = 6$ donors in each structure.

Why not the others: (A) changes relative ligand positions rather than donor identity; (B) would exchange a ligand with a counterion, which has not happened; (D) mistakes donor choice for mirror chirality and counts a nonbonded atom as a donor.

Key point: Compare donor atom, sphere boundary and spatial arrangement separately.

Q18 / D hydrogen transition wavelength

Answer: (D) 5R/36.

Given: 1/λ=R(1/nlow²−1/nhigh²).

Formula/Principle: Use 1/2²−1/3²=1/4−1/9.

Step 1: Difference 5/36.

Worked derivation:

$$1/\lambda = R(1/2^2 - 1/3^2) = R(1/4 - 1/9) = 5R/36.$$

Why: The lower final energy level has the larger inverse-square factor, making the emitted photon energy positive. Interchanging the level indices would give a negative inverse wavelength.

Why not the others: (A) subtracts denominators; (B) uses high state only; (C) uses low state only.

Key point: Subtract reciprocal squares in final-first order.

Q19 / A electrophile nucleophile selectivity

Answer: (A) Alkene π pair attacks electrophilic bromine.

Why: The electron-rich alkene polarises Br_2 and donates its π pair toward the electrophilic bromine end, beginning addition across C=C. Under the stated ordinary bromination conditions, a separate nonconjugated ketone is not the unsaturation consumed by this one-equivalent alkene test.

Why not the others: (B) has an anion acting as an electrophile; (C) both selects the wrong bond and reverses charge roles; (D) makes the bromide anion accept an electron pair and imagines oxygen leaving.

Key point: Name the electron donor and electron acceptor before choosing the reacting bond.

Q20 / A addition regioselectivity

Answer: (A) 2-bromopropane.

Why: Protonation that leaves a secondary carbocation at the middle carbon is favoured over a primary carbocation; bromide then attacks that centre. The peroxide-free premise excludes the radical anti-Markovnikov route associated with HBr under initiator conditions.

Why not the others: (B) uses less stable primary intermediate; (C) invents second Br; (D) removes bromine altogether.

Key point: Identify reaction conditions before regioselectivity.

Q21 / C SN1 versus SN2 outcome

Answer: (C) Substrate-controlled first-order rate.

Why: The slow heterolytic ionisation creates a carbocation before nucleophile capture, making the rate primarily proportional to alkyl-halide concentration. Competing elimination can occur from that same intermediate; changing nucleophile concentration need not control the ionisation step.

Why not the others: (A) suggests backside SN2; (B) implies bimolecular rate; (D) contradicts stable tertiary cation.

Key point: Diagnose rate-determining ionisation.

Q22 / D alcohol oxidation selectivity

Answer: (D) Primary alcohol.

Why: A primary alcohol first oxidises to an aldehyde, which reduces Tollens reagent and can oxidise further to a carboxylic acid. Secondary alcohol yields a ketone under ordinary oxidation, while a tertiary alcohol lacks the required carbinol hydrogen for this sequence.

Why not the others: (A) resists ordinary oxidation; (B) is not the stated alcohol class; (C) forms a ketone.

Key point: Follow the sequence through the carbonyl intermediate.

Q23 / A esterification equilibrium

Answer: (A) Equilibrium shifts to replace removed product.

Why: Esterification is reversible, so lowering water activity drives further forward reaction by the reaction quotient. The equilibrium constant at unchanged temperature is not itself increased; the mixture moves toward the same K at a new composition.

Why not the others: (B) misunderstands product role; (C) invents reduction; (D) changes reaction class.

Key point: Distinguish shifting composition from changing K.

Q24 / B Aniline protection and controlled ring substitution

Answer: (B) Temporarily moderate activation while preserving ortho/para direction.

Step 1: Acetylation changes $-NH_2$ into $-NHCOCH_3$; the nitrogen lone pair is shared partly with the carbonyl.

Step 2: Reduced donation into the ring moderates the multiple substitution characteristic of unprotected aniline, while the substituent remains ortho/para directing. The controlled conditions and steric preference favour the para product.

Step 3: Hydrolysis removes the acetyl group and regenerates NH_2 without relocating the ring bromine.

Why: Protection changes the reactivity during the substitution step and is then deliberately reversed. It does not permanently convert the substituent into a meta director or replace nitrogen with a new atom.

Why not the others: (A) confuses reduced activation with a change to meta direction; (C) reverses the effect of amide delocalisation; (D) contradicts the nitrogen atom retained throughout the displayed structures.

Key point: Follow both electronic effect and atom connectivity through protection and deprotection.

Q25 / C van Hoff factor inference

Answer: (C) 2.

Given: ΔTf=0.28 K, Kfm=0.14 K.

Formula/Principle: i=ΔTf/(Kfm).

Step 1: 0.28/0.14=2.

Worked derivation:

$$i = \frac{\Delta T_f}{K_f m} = \frac{0.28}{1.40(0.10)} = 2.$$

Why: The measured colligative effect is twice the molecular no-dissociation reference at equal molality. This effective particle factor does not by itself identify the salt formula or prove ideal complete dissociation.

Why not the others: (A) ignores enhancement; (B) underestimates; (D) overestimates.

Key point: Infer effective particle count from measured-to-ideal ratio.

Q26 / D concentration-cell potential

Answer: (D) Dilute side oxidises.

Why: Oxidation $M \rightarrow M^{2+}$ adds ions to the dilute half-cell, while reduction removes them from the concentrated one, tending toward equal activities. Identical standard potentials do not make the cell voltage zero when concentrations differ.

Why not the others: (A) increases disparity; (B) ignores Nernst concentration term; (C) violates a single net cell direction.

Key point: Concentration gradient can drive a cell.

Q27 / A rate graph and activation barrier

Answer: (A) X twice Y.

Given: slope=−Ea/R.

Formula/Principle: Ratio EaX/EaY=6000/3000.

Step 1: Ratio 2.

Worked derivation:

$$\text{slope} = -E_a/R, \quad \frac{E_{a,X}}{E_{a,Y}} = \frac{6000}{3000} = 2.$$

Why: The plot's horizontal variable is reciprocal temperature, so a more negative slope has greater magnitude and larger activation barrier. Equal intercepts fix pre-exponential factors but are unnecessary to infer the energy ratio.

Why not the others: (B) reverses slopes; (C) ignores magnitude; (D) overlooks Arrhenius form.

Key point: Use absolute slope for barrier.

Q28 / B Bomb-calorimeter calibration and internal-energy-to-enthalpy conversion

Answer: (B) Approximately $-152.5\ \text{kJ mol}^{-1}$.

Given: The same calibrated calorimeter receives the combustion heat at fixed volume; gases are treated ideally for the enthalpy conversion.

Formula/Principle:

$$q_{\text{cal}} = C_{\text{cal}}\Delta T, \quad \Delta U_m = -q_{\text{cal}}/n, \quad \Delta H_m = \Delta U_m + \Delta n_g RT.$$

Step 1:

$$C_{\text{cal}} = 1000/2.00 = 500\ \text{J K}^{-1}.$$

Step 2:

$$q_{\text{cal}} = 500(3.00) = 1500\ \text{J}, \quad \Delta U_m = -1500/0.0100 = -150000\ \text{J mol}^{-1}.$$

Step 3:

$$\Delta H_m = -150000 - 1(8.30)(300) = -152490\ \text{J mol}^{-1} \approx -152.5\ \text{kJ mol}^{-1}.$$

Why: A rigid calorimeter measures the reaction's constant-volume heat and therefore its internal-energy change. The gas mole change must be included when converting that quantity to enthalpy; heat gained by the calorimeter has the opposite sign to reaction heat.

Why not the others: (A) stops at internal energy; (C) reverses the reaction-heat sign; (D) uses the opposite sign for the gas correction.

Key point: Calibrate first, determine constant-volume heat, then convert internal energy to enthalpy.

Q29 / C weak-acid dissociation

Answer: (C) 10^{-2}.

Given: $Ka \approx C\alpha^2$ for $\alpha \ll 1$.

Formula/Principle: $\alpha \approx \sqrt{(Ka/C)}$.

Step 1: $\sqrt{(10^{-5}/10^{-1})} = \sqrt{10^{-4}} = 10^{-2}$.

Worked derivation:

$$\alpha \simeq \sqrt{K_a/C} = \sqrt{10^{-5}/10^{-1}} = 10^{-2}, \quad \alpha = 0.01 \ll 1.$$

Why: One dissociation produces one H^+ and one A^-, giving a squared ionisation fraction in the mass-action expression. The resulting 1% ionisation is small enough to support the approximation.

Why not the others: (A) takes square root wrongly; (B) overestimates; (D) uses Ka/C without root.

Key point: Check the approximation after solving.

Q30 / D variable oxidation-state stability

Answer: (D) Lower Mn state in acid.

Why: Acid supplies protons needed to balance deeper permanganate reduction to Mn^{2+}, and its redox potential is correspondingly stronger in that medium. Basic conditions often favour MnO_2 or manganate pathways depending on conditions, rather than the same five-electron acidic half-reaction.

Why not the others: (A) denies redox; (B) ignores Mn; (C) impossible oxidation beyond +7 here.

Key point: Redox products depend on pH and medium.

Mathematics

Q31 / D Chord endpoints determine two tangents and their intersection

Answer: (D) 3/2.

Given: The circle has centre $C = (3, 0)$ and radius 3.

Formula/Principle: Find the two contact points, then use the radius direction as the normal to each tangent.

Step 1:

$$(1-3)^2 + y^2 = 9 \Rightarrow P = (1, \sqrt{5}), \quad Q = (1, -\sqrt{5}).$$

Step 2: The tangent at P is

$$(-2)(x-1) + \sqrt{5}(y - \sqrt{5}) = 0 \Rightarrow -2x + \sqrt{5}y = 3.$$

Step 3: At Q the tangent is $-2x - \sqrt{5}y = 3$. Solving the pair gives $y_T = 0,\ x_T = -3/2$.

Step 4:

$$OT = \sqrt{(-3/2)^2} = 3/2.$$

Worked derivation:

$$T = (-3/2, 0), \qquad OT = 3/2.$$

Why: The chord endpoints are needed as contact points, but the requested length is not the chord length. Symmetry places the tangent intersection on the x-axis, while the tangent equations determine which side of the origin it occupies.

Why not the others: (A) gives distance CT; (B) gives the radius; (C) gives the diameter.

Key point: Distinguish centre distances, chord lengths and tangent-intersection coordinates.

Q32 / A ellipse focus-directrix

Answer: (A) $x^2/25+y^2/16=1$.

Given: $c=3$ and directrix distance $a/e=a^2/c=25/3$.

Formula/Principle: $a^2=25$, $b^2=a^2-c^2=16$.

Step 1: Major axis x; substitute denominators.

Worked derivation:

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, \quad a^2=25,\ b^2=25-9=16.$$

Why: The focus coordinate and directrix jointly determine a^2, not just a; the minor-axis denominator follows from $c^2=a^2-b^2$. The positive x direction fixes major-axis orientation.

Why not the others: (B) swaps axes; (C) uses c^2 as a^2; (D) uses c^2 as b^2.

Key point: Combine focus and directrix information.

Q33 / B area between curves

Answer: (B) 32/3.

Given: Intercepts $x=-2$ and 2.

Formula/Principle: Integrate nonnegative curve over interval.

Step 1: $[4x-x^3/3]_{-2}^{2}=32/3$.

Worked derivation:

$$A=\int_{-2}^{2}(4-x^2)dx=2\left[4x-\frac{x^3}{3}\right]_0^2=\frac{32}{3}.$$

Why: Symmetry doubles the right-half area and the curve remains above the x-axis between roots. Integrating only 0 to 2 would give half the region.

Why not the others: (A) halves; (C) uses bounding rectangle partly; (D) takes height only.

Key point: Locate both crossings and exploit symmetry.

Q34 / C substitution and bounds

Answer: (C) 1/2.

Given: Put $u=\ln x$, $du=dx/x$, bounds 0 to 1.

Formula/Principle: Integrate u on transformed interval.

Step 1: $[u^2/2]_0^1=1/2$.

Worked derivation:

$$\int_1^e \frac{\ln x}{x}dx=\int_0^1 u\,du=\frac{1}{2}.$$

Why: The differential dx/x matches the logarithmic derivative, so substituting changes both integrand and endpoints. Treating ln x as constant would not preserve the varying area.

Why not the others: (A) omits integration factor; (B) halves twice; (D) doubles result.

Key point: Update bounds for definite substitution.

Q35 / D monotonicity interval

Answer: (D) (0,3).

Given: f′=1−9/x².

Formula/Principle: On positive domain, f′<0 iff x²<9.

Step 1: 0<x<3.

Worked derivation:

$$f'(x) = \frac{x^2 - 9}{x^2} < 0 \iff 0 < x < 3.$$

Why: The denominator is positive on the given domain, so the sign is controlled by x²−9. The derivative changes sign at 3, giving a minimum there rather than continued decrease.

Why not the others: (A) is increasing; (B) crosses the minimum; (C) includes an increasing segment.

Key point: Apply derivative sign with domain.

Q36 / A injection versus surjection

Answer: (A) Onto but not one-to-one.

Worked derivation:

$$x + 1/x \geq 2; \quad f(x) = f(1/x), \quad x \neq 1.$$

Why: AM–GM supplies the exact range [2,∞), matching the codomain, while reciprocal positive inputs collide. The codomain must be read as stated; replacing it with all real numbers would alter the surjectivity verdict.

Why not the others: (B) reverses both; (C) ignores collisions; (D) ignores complete range.

Key point: Determine range and collisions independently.

Q37 / B adjoint identity

Answer: (B) 4.

Worked derivation:

$$A \operatorname{adj} A = (\det A) I \Rightarrow \det A \det(\operatorname{adj} A) = (\det A)^3 \Rightarrow \det(\operatorname{adj} A) = 2^2 = 4.$$

Why: Determinants multiply across the adjoint identity, and the scalar 2 multiplies all three diagonal entries of I. Canceling one factor of det A leaves its square in three dimensions.

Why not the others: (A) misses exponent; (C) retains cube; (D) inverts it.

Key point: Adj determinant scales as (det A)^(n−1).

Q38 / C independent versus exclusive events

Answer: (C) No.

Worked derivation:

$$P(A \cap B) = 0 \neq (1/2)(1/2) = 1/4.$$

Why: Independence demands the intersection probability equal the product of marginals, whereas mutually exclusive positive-probability events never co-occur. Only if at least one event has zero probability can both properties coexist.

Why not the others: (A) conflates distinct properties; (B) overrestricts the exception; (D) sample-space size is irrelevant.

Key point: Compare intersection with product.

Q39 / D periodic solution count

Answer: (D) 3.

Given: cosθ=1 iff θ=2πk.

Formula/Principle: 3x=2πk; 0≤x<2π gives k=0,1,2.

Step 1: Three distinct solutions.

Worked derivation:

$$x \in \{0, 2\pi/3, 4\pi/3\}, \quad N = 3.$$

Why: The half-open interval excludes x=2π, which would duplicate the same phase as zero but be another real endpoint on a closed interval. The triple argument completes three cosine cycles.

Why not the others: (A) omits one period; (B) includes excluded endpoint; (C) counts ± solutions for each peak.

Key point: Respect half-open endpoints.

Q40 / A Parallel-line condition and distance from a plane

Answer: (A) $3;\ \sqrt{6}/2$.

Given: The plane normal is $\mathbf{n} = (2, -1, 1)$, with magnitude $\sqrt{6}$.

Formula/Principle: A parallel line has direction perpendicular to the plane normal; all its points then have the same plane distance.

Step 1:

$$(1, k, 1) \cdot (2, -1, 1) = 3 - k = 0 \Rightarrow k = 3.$$

Step 2: Use the given point on the line:

$$d = \frac{|2(1) - 0 + 0 - 5|}{\sqrt{2^2 + (-1)^2 + 1^2}} = \frac{3}{\sqrt{6}} = \frac{\sqrt{6}}{2}.$$

Step 3: The point is not in the plane, so the parallel line is distinct from the plane and the distance is positive.

Worked derivation:

$$k = 3, \qquad d = \frac{\sqrt{6}}{2}.$$

Why: Parallelism fixes the direction but does not by itself imply that the line lies in the plane. The separate point-to-plane calculation supplies that missing positional information.

Why not the others: (B) omits normalisation by the normal length; (C) loses the sign in the dot product; (D) fails the parallelism condition.

Key point: Check direction and position as separate line-plane conditions.

Q41 / B collinearity test

Answer: (B) Yes; 2:3.

Given: B−A=(2,2,2); C−B=(3,3,3).

Formula/Principle: Parallel difference vectors establish one straight line; length ratio is 2√3:3√3.

Step 1: Yes, 2:3.

Worked derivation:

$$\vec{AB} = 2(1, 1, 1), \quad \vec{BC} = 3(1, 1, 1) \Rightarrow AB : BC = 2 : 3.$$

Why: The two consecutive displacements are positive scalar multiples, so B lies between A and C on the same line. Comparing coordinate differences rather than distances alone establishes collinearity.

Why not the others: (A) ignores parallelism; (C) reverses lengths; (D) misstates both.

Key point: Compare direction and magnitude.

Q42 / C Unequal complex-distance ratio and a circle

Answer: (C) Centre $(-5/3, 0)$, radius $4/3$.

Given: Both distances are nonnegative, so squaring their equality is reversible.

Formula/Principle: Expand the squared distance equation and complete the square.

Step 1:

$$(x-1)^2 + y^2 = 4[(x+1)^2 + y^2].$$

Step 2:

$$3x^2 + 10x + 3y^2 + 3 = 0.$$

Step 3:

$$\left(x + \frac{5}{3}\right)^2 + y^2 = \frac{16}{9}.$$

Worked derivation:

$$\text{centre} = (-5/3, 0), \qquad r = \sqrt{16/9} = 4/3.$$

Why: A distance ratio different from one produces a circle rather than the perpendicular bisector associated with equal distances. Reading the centre requires the opposite sign of the completed-square shift.

Why not the others: (A) reverses the centre sign; (B) takes the wrong square root; (D) ignores the unequal distance ratio.

Key point: Preserve the squared distance ratio before completing the square.

Q43 / D Two roots in distinct strict intervals

Answer: (D) $3/2 < k < 11/6$.

Given: The quadratic is continuous and opens upward; the requested two intervals do not overlap.

Formula/Principle: A sign change in each interval guarantees a root there; a quadratic can have only two roots in total.

Step 1:

$$f(0) = 2 > 0, \quad f(1) = 3 - 2k < 0 \Rightarrow k > 3/2.$$

Step 2:

$$f(2) = 6 - 4k < 0 \Rightarrow k > 3/2, \quad f(3) = 11 - 6k > 0 \Rightarrow k < 11/6.$$

Step 3: Combining the strict inequalities gives $(3/2, 11/6)$, which supplies one sign-changing root in each required interval.

Step 4: Conversely, the specified root placement puts 1 and 2 between the roots and 0 and 3 outside them, requiring exactly these signs. The boundary values are excluded because a root reaches an interval endpoint.

Worked derivation:

$$f(1) < 0,\ f(2) < 0,\ f(3) > 0 \iff 3/2 < k < 11/6.$$

Why: Reality alone would not locate the roots in the two separate open intervals. Checking both interval boundaries supplies a necessary and sufficient condition and excludes endpoint roots without relying on a picture.

Why not the others: (A) permits boundary roots; (B) omits the upper interval bound; (C) omits the lower interval bound.

Key point: Use endpoint signs to enforce two disjoint strict root intervals.

Q44 / A neighboring-coefficient identity

Answer: (A) 112.

Given: Coefficients are C(8,3) and C(8,5), equal by symmetry.

Formula/Principle: C(8,3)=56.

Step 1: 56+56=112.

Worked derivation:

$$\binom{8}{3}+\binom{8}{5}=2\binom{8}{3}=112.$$

Why: Complementary binomial indices give equal coefficients; the sum therefore doubles one of them. The central coefficient C(8,4)=70 is a different term.

Why not the others: (B) takes one; (C) takes central; (D) chooses wrong binomial index.

Key point: Use symmetry of neighboring coefficients.

Q45 / B Inserting arithmetic means

Answer: (B) 30.

Given: The progression has first term 4 and fifth term 16.

Formula/Principle: Four equal differences span the endpoints.

Step 1:

$$d=\frac{16-4}{4}=3.$$

Step 2: The inserted terms are $7,10,13$, so their sum is 30.

Worked derivation:

$$7+10+13=30=3\left(\frac{4+16}{2}\right).$$

Why: Inserting three means creates four gaps, not three. The symmetry about the middle value ten also checks the sum independently.

Why not the others: (A) adds only the endpoints; (C) uses the wrong number of gaps; (D) includes an extra copy of the full progression's sum.

Key point: Count gaps when inserting arithmetic means.

Physics

Q46 / C relative motion with shared origin

Answer: (C) 8 s.

Given: xA=12t; xB=½(3)t².

Formula/Principle: Equate positions and exclude shared starting instant t=0.

Step 1: 12t=1.5t² gives t=8 s.

Worked derivation:

$$t(12-1.5t)=0\Rightarrow t=8\ \text{s}\quad (t>0).$$

Why: Equal speed would happen at a different time, since B's velocity changes while A's stays fixed. The word next removes the trivial common origin.

Why not the others: (A) is equal-speed time; (B) mixes speed and displacement; (D) overextends.

Key point: Equate positions for a meeting.

Q47 / D error propagation

Answer: (D) 2%.

Given: Each side fractional uncertainty is 1%.

Formula/Principle: For product, maximum fractional errors add.

Step 1: 1%+1%=2%.

Worked derivation:

$$\frac{\Delta A}{A} \lesssim \frac{0.1}{10.0} + \frac{0.2}{20.0} = 0.02 = 2\%.$$

Why: In a product the leading worst-case relative changes add; errors do not add as absolute centimetres to an area. The estimate is a maximum bound, not a statistical standard uncertainty.

Why not the others: (A) keeps one contribution; (B) averages; (C) doubles twice.

Key point: Add relative errors for a product.

Q48 / A Free-surface slope from effective gravity

Answer: (A) Left side higher; angle tangent a/g.

Given: Choose x rightward and z upward; in the tank frame the effective acceleration is $(-a, -g)$.

Formula/Principle: Hydrostatic pressure satisfies $dp = -\rho a\, dx - \rho g\, dz$; pressure is constant along the free surface.

Step 1: Set $dp = 0$ on that surface:

$$a\, dx + g\, dz = 0.$$

Step 2:

$$\frac{dz}{dx} = -\frac{a}{g} < 0.$$

Height therefore falls toward the right.

Step 3: The positive inclination magnitude satisfies $\tan\theta = |dz/dx| = a/g$.

Worked derivation:

$$z(x) = z(0) - \frac{a}{g}x, \qquad \tan\theta = \frac{a}{g}.$$

Why: The noninertial-frame acceleration has a horizontal component opposite the tank's acceleration. A constant-pressure surface must be perpendicular to effective gravity, so the surface is not horizontal in that frame.

Why not the others: (B) reverses the height gradient; (C) inverts the slope ratio; (D) makes both errors.

Key point: Use effective gravity and constant pressure rather than guessing a sloshing direction.

Q49 / B drilled-body inertia

Answer: (B) 5mR²/8.

Given: An annulus with inner radius r=R/2 and outer radius R.

Formula/Principle: I=½m(R²+r²).

Step 1: ½m(1+1/4)R²=5mR²/8.

Worked derivation:

$$I = \frac{m}{2}\left(R^2 + \frac{R^2}{4}\right) = \frac{5}{8}mR^2.$$

Why: The removed mass changes both total mass and its radial distribution; expressing the answer in remaining mass avoids treating it as the original full disk. The annulus formula follows by subtracting coaxial disk inertias.

Why not the others: (A) uses full disk; (C) uses wrong inner radius; (D) overweights rim.

Key point: Distinguish original and remaining mass.

Q50 / C Cascaded cyclic heat-engine balances with an external load

Answer: (C) Useful efficiency 55%; final rejected heat 320 J.

Given: Each engine is cyclic, and the entire first rejection becomes the second heat input.

Formula/Principle: For each engine $W = \eta Q_{\text{in}}$ and $Q_{\text{out}} = Q_{\text{in}} - W$.

Step 1:

$$W_1 = 0.40(800) = 320, \quad Q_{12} = 800 - 320 = 480 \text{ J}.$$

Step 2:

$$W_2 = (1/3)(480) = 160, \quad Q_{\text{out},2} = 480 - 160 = 320 \text{ J}.$$

Step 3:

$$W_{\text{useful}} = 320 + 160 - 40 = 440 \text{ J}.$$

Step 4:

$$\eta_{\text{useful}} = 440/800 = 0.55 = 55\%.$$

Worked derivation:

$$800 = 440 + 40 + 320 \text{ J}.$$

Why: The second efficiency acts on the first engine's rejected heat, not on the original 800 J again. The parasitic load reduces useful work but, by stipulation, does not change either engine's own heat balance.

Why not the others: (A) reports gross rather than useful efficiency; (B) reports the inter-engine heat transfer as final rejection; (D) ignores both the second engine's contribution and the external loss accounting.

Key point: Draw separate cyclic energy balances before combining a cascade's useful output.

Q51 / B spring-combination period

Answer: (B) 1/2.

Given: Parallel spring stiffness k_eff=k+3k=4k.

Formula/Principle: T=2π√(m/k_eff).

Step 1: Tparallel/Tk=√(k/4k)=1/2.

Worked derivation:

$$\frac{T_{\text{parallel}}}{T_k} = \sqrt{\frac{k}{k+3k}} = \frac{1}{2}.$$

Why: Both springs experience the same displacement and their restoring forces add, making stiffness fourfold. Period falls with the square root, not linearly.

Why not the others: (A) inverts; (C) omits square root; (D) uses 3k alone.

Key point: Combine stiffness before period.

Q52 / C interference with source phase

Answer: (C) Zero.

Worked derivation:

$$I = 4I_0 \cos^2(\Delta\phi/2) = 4I_0 \cos^2(\pi/2) = 0.$$

Why: Equal paths contribute no extra propagation phase, so the imposed source phase difference remains π and equal amplitudes cancel. Coherence keeps that phase relation stable over observation time.

Why not the others: (A) is in-phase maximum; (B) adds intensities without interference; (D) ignores second source.

Key point: Include initial phase as well as path phase.

Q53 / D dipole torque and energy

Answer: (D) +pE.

Given: U=−pE cosθ.

Formula/Principle: Uinitial=−pE; Ufinal=0.

Step 1: ΔU=0−(−pE)=pE.

Worked derivation:

$$\Delta U = -pE\cos(\pi/2) + pE\cos 0 = pE.$$

Why: Parallel orientation is the lower-energy state; rotating away against the aligning torque requires positive work stored as potential energy. This is an energy difference, not the torque at one endpoint.

Why not the others: (A) reverses sign; (B) evaluates final alone; (C) uses antiparallel final.

Key point: Subtract initial from final energy.

Q54 / A drift versus current density

Answer: (A) 2×10⁶ A m⁻².

Given: Area=2×10⁻⁶ m².

Formula/Principle: J=I/A.

Step 1: 4/(2×10⁻⁶)=2×10⁶.

Worked derivation:

$$J = \frac{4\ \text{A}}{2\times10^{-6}\ \text{m}^2} = 2\times10^{6}\ \text{A m}^{-2}.$$

Why: Millimetre squared converts by 10^{-6} to metre squared; treating mm as a linear factor would introduce a millionfold error. Drift speed also needs carrier density and charge, neither given here.

Why not the others: (B) ignores area conversion; (C) multiplies by area; (D) doubles instead of divides.

Key point: Convert area units before current density.

Q55 / D Equal activities and separate parent decay populations

Answer: (D) 525 daughter atoms.

Given: $\lambda_P = 2\lambda_Q$ and $N_{P0} + N_{Q0} = 900$.

Formula/Principle: Equal activity means $\lambda_P N_{P0} = \lambda_Q N_{Q0}$, not equal initial populations.

Step 1:

$$N_{Q0} = 2N_{P0} \Rightarrow N_{P0} = 300, \quad N_{Q0} = 600.$$

Step 2: After 4 h,

$$N_P = 300(1/2)^2 = 75, \qquad N_Q = 600(1/2) = 300.$$

Step 3:

$$N_D = 900 - (75 + 300) = 525.$$

Worked derivation:

$$N_D = 300(1 - 1/4) + 600(1 - 1/2) = 225 + 300 = 525.$$

Why: The shorter-lived parent needs fewer atoms to have the same initial decay activity. Stable daughters accumulate from both independent parent populations, so atom conservation can be applied after calculating each surviving population separately.

Why not the others: (A) reports surviving parents; (B) assumes an unsupported one-half decay of the whole mixture; (C) ignores the actual two-component survival calculation.

Key point: Translate equal activity into a population ratio before applying separate half-lives.

Q56 / B mutual induction sign

Answer: (B) Away from observer.

Worked derivation:

$$\mathcal{E}_2 = -M\, dI_1/dt; \quad d\Phi_{1\to 2}/dt > 0 \Rightarrow B_{\text{induced}} \text{ opposes positive flux.}$$

Why: The primary's outward flux is increasing, so Lenz's law calls for an inward secondary field. Mutual induction requires changing magnetic linkage, not electrical contact between loops.

Why not the others: (A) reinforces change; (C) denies induction; (D) invents alternating drive.

Key point: Oppose flux change before assigning current.

Q57 / A critical angle through layers

Answer: (A) Total internal reflection.

Worked derivation:

$$\theta_c = \sin^{-1}(1/1.5) \approx 41.8^\circ, \quad 50^\circ > \theta_c.$$

Why: Incidence from higher to lower index above the critical angle cannot satisfy a real transmitted Snell angle. Reflection remains in glass, though a near-interface evanescent field can exist without an ordinary transmitted ray.

Why not the others: (B),(C) require a real refracted ray; (D) is not implied.

Key point: Compare incidence with critical angle.

Q58 / B phasor phase and impedance

Answer: (B) 4/3.

Given: Net reactance positive (inductive).

Formula/Principle: tanφ=(XL−XC)/R.

Step 1: tanφ=4/3; current lags voltage.

Worked derivation:

$$\tan\phi = \frac{X_L - X_C}{R} = \frac{4}{3}.$$

Why: Resistance is the in-phase component and net reactance the quadrature component of impedance. Their ratio sets phase; adding their magnitudes arithmetically would confuse phase with impedance.

Why not the others: (A) inverts; (C) adds; (D) uses total sum.

Key point: Keep phasor components perpendicular.

Q59 / C half-wave output power

Answer: (C) $V_0^2/(4R)$.

Given: Power is $V_0^2\sin^2\omega t/R$ for positive half-cycle, zero for negative half.

Formula/Principle: Full-cycle average of $\sin^2$ on one half is 1/4.

Step 1: $P_{avg}=V_0^2/(4R)$.

Worked derivation:

$$P_{\text{avg}} = \frac{1}{2\pi}\int_0^{\pi} \frac{V_0^2 \sin^2\theta}{R} d\theta = \frac{V_0^2}{4R}.$$

Why: Power depends on the square of instantaneous voltage, so squaring average voltage is not equivalent to averaging power. The diode removes the negative half-cycle rather than reversing it.

Why not the others: (A) uses peak; (B) uses full sine; (D) halves twice.

Key point: Average instantaneous power over entire period.

Q60 / D Fringe spacing and tracked phase-sheet displacement

Answer: (D) New width $\beta/2$; thickness 3λ.

Given: D and wavelength in air are unchanged; the specified tracked fringe is the zero-total-path-difference fringe, not just the nearest bright fringe to the axis.

Formula/Principle: A uniform sheet adds path difference $\Delta = (n-1)t$; central-fringe displacement magnitude is $y_0 = \Delta D/d_{\text{new}}$.

Step 1:

$$\beta_{\text{new}} = \frac{\lambda D}{2d} = \frac{\beta}{2}.$$

Step 2:

$$\frac{y_0}{\beta} = \frac{\Delta D/(2d)}{\lambda D/d} = \frac{\Delta}{2\lambda} = \frac{3}{4}.$$

Step 3:

$$\Delta = \frac{3}{2}\lambda, \quad t = \frac{\Delta}{1.50 - 1} = \frac{(3/2)\lambda}{1/2} = 3\lambda.$$

Worked derivation:

$$\beta_{\text{new}} = \beta/2, \qquad t = 3\lambda.$$

Why: The sheet translates the pattern but does not alter its spacing; the changed slit separation affects both width and the conversion from path difference to displacement. Tracking the specified zero-path fringe avoids an ambiguity by whole fringe orders when interpreting the shift.

Why not the others: (A) reverses the separation scaling; (B) reports the added path as thickness; (C) inserts an extra factor of two in the sheet correction.

Key point: Recompute geometric spacing before converting a measured phase shift to thickness.